

SAT: Scalar-Angular-Theory – A 4D Tensorial–Discrete Twist Framework under the Lean 4 Formalism

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We introduce the SAT framework, a four-dimensional tensorial and discrete twist integration theory expressed within the Lean 4 formal environment. The formulation unifies topological invariants, mass spectra, and 4D field interactions through a hybrid continuous–discrete Lagrangian density. We present the explicit Lagrangian structure, describe its physical sectors, and discuss preliminary implications for quantization and topological mass suppression.

I. INTRODUCTION

The SAT Self-Adjusting Tensor (SAT) framework proposes a field-theoretic structure that bridges continuous tensor dynamics and discrete topological information. It extends conventional 4D field models by integrating a *twist charge* τ sector, encoding winding, linking, and writhe numbers. These invariants enter both the mass spectrum and the topological action, resulting in a discretely modulated continuous Lagrangian.

II. LAGRANGIAN FORMULATION

The total Lagrangian density is defined as

$$\mathcal{L}_{\text{total}} = \underbrace{(\theta_\mu)^2}_{\text{scalar potential}} + \underbrace{(u_\mu)^2}_{\text{vector kinetic term}} + \underbrace{(J_\mu^\mu)^2}_{\text{filament current energy}} + \underbrace{\mathcal{S}_{\text{BF}}(u, \theta, \tau)}_{\text{BF topological term}} + \underbrace{\mathcal{K}_{\text{fil}}(J, \tau)}_{\text{filament kernel}}. \quad (1)$$

Each component corresponds to a physically distinct sector:

- $\theta_\mu(x)$ — scalar field amplitude in 4D spacetime.
- $u_\mu(x)$ — vector (strain-like) field encoding elastic deformations.
- $J^{\mu\nu}(x)$ — antisymmetric filament current tensor.
- $\tau = (L_{\text{wind}}, L_{\text{link}}, W_{\text{writhe}})$ — discrete twist charge describing topological embedding.

A. Topological Sector and BF Action

The BF-type coupling introduces the discrete twist charge τ through

$$\mathcal{S}_{\text{BF}}(u, \theta, \tau) = \int_0^1 dx \left[B_\mu(x) \nabla^\mu A(x) + Q(\tau) \right], \quad (2)$$

where $Q(\tau) = L_{\text{wind}} + L_{\text{link}} + W_{\text{writhe}}$ is the total twist number.

B. Filament Kernel Contribution

A damping kernel suppresses the topological mass via

$$\mathcal{K}_{\text{fil}}(J, \tau) = \int_0^1 dx J_{00}(x) e^{-Q(\tau)}. \quad (3)$$

C. Mass Spectrum

The effective mass of a topologically charged mode is given by

$$m_{\text{eff}} = m_{\text{topo}} + m_{\text{ind}}, \quad (4)$$

$$m_{\text{topo}} = \frac{m_0}{Q(\tau)}, \quad (5)$$

$$m_{\text{ind}} = \frac{g_1^2}{M^2} E_{\text{overlap}}, \quad (6)$$

with g_1 a coupling constant, M a mass scale, and E_{overlap} an energy overlap integral parameterizing interaction strength.

D. Euler–Lagrange Structure

The dynamical evolution follows from the Euler–Lagrange equations for the primary fields:

$$(i) \quad \nabla_\mu \nabla^\mu \theta + 3 \sin(3\theta) = 0, \quad (7)$$

$$(ii) \quad \nabla_\mu \nabla^\mu u^\nu + \Lambda u^\nu + \theta = 0, \quad (8)$$

$$(iii) \quad \nabla_\mu J^{\mu\nu} = 0, \quad (9)$$

where Λ is a Lagrange-type multiplier coupling scalar and vector components.

III. DISCUSSION

The SAT framework encapsulates a rich interplay between continuous fields and discrete topology. The integral structure over τ permits quantized deformation modes and potential realizations of discrete topological sectors in elastic or quantum condensate systems.

Canonical quantization proceeds through symbolic commutators,

$$[\xi(\lambda), \pi(\lambda')] = i\pi \delta(\lambda - \lambda'), \quad (10)$$

indicating a standard quantization condition over the continuous sector.

field space and a topological twist charge. The resulting Lagrangian structure naturally blends topological invariants with dynamic field degrees of freedom, yielding a potentially rich landscape for emergent mass spectra and geometric quantization effects.

IV. CONCLUSION

We have formulated the SAT theory as a discrete-continuous hybrid model incorporating a 4D tensorial
